

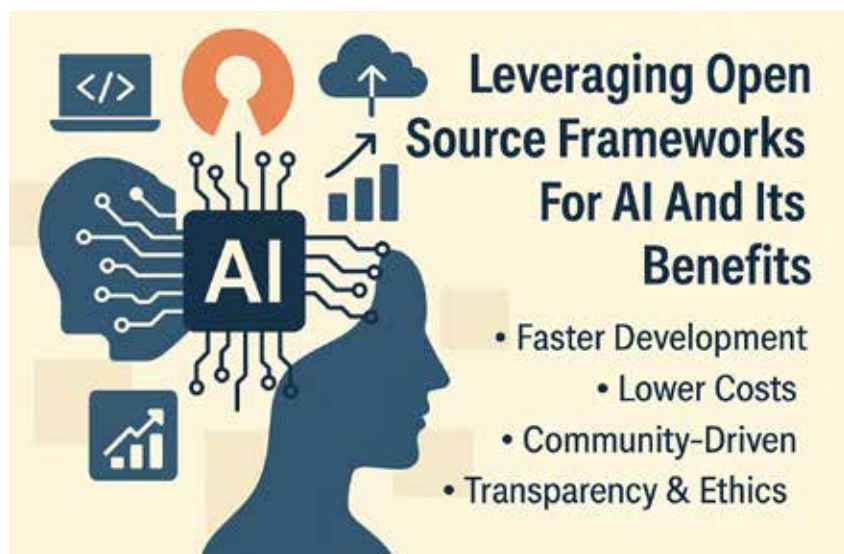


Figure 2: Open source AI benefits (Credit: ChatGPT)

A comparative overview of widely-used open source frameworks shows how each serves specific needs:

Framework	Best use cases
TensorFlow	Deep learning, ML model training
PyTorch	Research and prototyping, NLP, Vision
Hugging Face	NLP and Transformer models
LangChain	LLM-based workflow development, AI agents
ONNX	Model interoperability and porting

pool of developers and researchers. This collaborative environment not only enhances the quality and performance of tools but also facilitates knowledge sharing. Developers can learn from others' work, incorporate new best practices, and contribute their own enhancements, thereby pushing the boundaries of what these frameworks can achieve.

Lastly, open source software brings transparency and ethical accountability to the forefront. Since the source code is openly available, organisations can thoroughly audit the codebase to ensure it aligns with their ethical standards and data governance policies. This visibility helps identify any potential security or bias issues, and allows developers to confidently build and deploy AI

solutions that are not only functional but also trustworthy.

Addressing the challenges

Despite the many advantages, open source AI frameworks do present certain hurdles. Here's how developers can tackle them:

Steep learning curve: Thoroughly explore documentation, community blogs, and GitHub repositories. Consider MOOCs and online tutorials to get up to speed.

Computational requirements: AI models demand high-performance computing. Leverage cloud platforms or AI-as-a-Service models to scale compute power as needed.

Data privacy concerns: Sensitive data exposure is a major issue. Adopt

federated learning approaches or use local AI models to train data privately before syncing with central servers.

Model bias: AI systems may reflect gender or racial biases. Using large, diverse datasets during training helps mitigate this issue.

For those inspired to contribute to open source AI, the journey begins with exploration. Projects like Mattermost, Llama models, and NVIDIA's BRUT N1 are fully open source. The Hugging Face ecosystem, in particular, offers a transparent dataset and model repository, enabling experimentation and contribution.

To get involved in open source AI projects, start by thoroughly reading the project documentation and contributor guides to understand the framework's structure, coding standards, and contribution workflow. Next, actively engage with the project maintainers by participating in discussions and raising relevant issues on GitHub. Once you have a clear idea or enhancement to propose, ensure it aligns with the project's goals and doesn't conflict with existing features. After discussing it with the maintainers and receiving their feedback or approval, you can implement the changes and submit a pull request for review and potential integration into the main codebase.

Open source AI isn't just about free tools—it's about democratising access to intelligence. Whether you're an independent developer or a CTO scaling enterprise systems, leveraging these frameworks can lead to faster innovation, reduced costs, and deeper community engagement. As AI evolves, open source will remain the engine powering its most inclusive and ethical advancements. **END** 🐙

This article is based on a tech talk by Abhishek Mankuskar, creator & lead, Social Winter Of Code, at AI DevCon 2025 in Bengaluru.

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